Campusmonk

Topic- Permutation & Combination -01

LIVE CLASS

Permutation

Combination

•1. How many batting orders are possible for the Indian cricket team if there is a squad of 15 to choose from such that Sachin Tendulkar is always chosen?

•2. In how many ways, can we select a team of 4 students from a given choice of 15 ?

Type : **Word Formation**

3. Find the number of words, with or without meaning, that can be formed with the letters of the word 'INDIA'.

4. Find the number of words, with or without meaning, that can be formed with the letters of the word 'SWIMMING?

Q5: In how many different ways can the letters of the word 'OPTICAL' be arranged so that the vowels always come together?

Q6: In how many different ways can the letters of the word 'LEADING' be arranged in such a way that the vowels always come together?

A. 360
B. 480
C. 720
D. 5040
E. None of these

Q7: How many different words can be formed with the letters of the word ‘SUPER’ such that the vowels always come together?

Q8:Find the number of different words that can be formed with the letters of the word ‘BUTTER’ so that the vowels are always together.

Q9: In how many ways can 10 examination papers be arranged so that the best and the worst papers never come together?

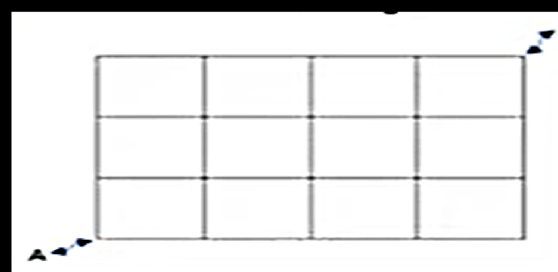
A. $8 \times 9!$
B. $8 \times 8!$
C. $7 \times 9!$
D. $9 \times 8!$

Q10: In how many different ways can the letters of the word 'DETAIL' be arranged in such a way that the vowels occupy only the odd positions?

A. 32
B. 48
C. 36
D. 60
E. 120

Q11: A person has to cover 6 meters using either a 1 meter step or a 2 meter step or a combination of both .Find in how many ways you can do it

Q12: Find the number of ways to reach from point A to C if the person has to take the shortest route and he can move in horizontal and vertical direction only



Q13: In how many ways can 8 Indians and, 4 American and 4 Englishmen can be seated in a row so that all person of the same nationality sit together?

A. 3! 4! 8! 4!
B. 3! 8!
C. 4! 4!
D. 8! 4! 4!

Q14: In how many ways can 12 examination papers be arranged so that the best and the worst papers never come together?

A. 10 × 11!
B. 8 × 8!
C. 7 × 9!
D. 9 × 8!

Number Formation :

Q15: How many numbers between 400 and 1000 can be formed with the digits 0,2,3,4,5,6, if no digit is repeated in the same number?

Q16: How many numbers between 300 and 3,000 can be formed with the digits 0, 1, 2, 3, 4, 5.

Q17: How many 3-digit numbers can be formed from the digits 2, 3, 5, 6, 7 and 9, which are divisible by 5 and none of the digits is repeated?

A. 5
B. 10
C. 15
D. 20

Q18: how many even four-digit numbers can be formed with the digit 0,1 ,2 ,3 ,4,5,6; no digit being used more than once?

Q19:MR.X has forgotten his friend's seven-digit telephone number. He remembers the following: the first three digits are either 635 or 674, the number is odd, and the number nine appears once. If X were to use a trial and error process to reach his friend, what is the number of trials he has to make before he can be certain to succeed?

A)1000
B)3402
C)2430
D)3006

Q20: In a birthday party, every person shakes hand with every other person. If there was a total of 28 handshakes in the party, how many persons were present in the party?

A. 8
B. 9
C. 6
D. 7